

The Bias-Variance Trade-off

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Where this fits: Pipeline map step 9 (Evaluate). See the [Course map](#).



- **Builds on:** Overfitting ([Note 61](#)); Underfitting ([Note 61](#)); Polynomial regression ([Note 61](#)).
- **Leads to:** Ridge regression ([Note 63](#)); Ensemble learning ([Note 101](#)); Bagging ([Note 105](#)).
- **Compare with:** K-nearest neighbours ([Note 6](#)).

1. Overview

Key point: A model's error on new data comes from two sources: bias (it is too simple to capture the pattern) and variance (it changes too much with the training data). Making one smaller usually makes the other larger.

The polynomial regression Note showed that a degree too low underfits and a degree too high overfits. This Note gives the two problems their standard names, **bias** and **variance**, and explains why reducing one tends to increase the other. This tension is the **bias-variance trade-off**, one of the central ideas of ML.

It is also a very common interview question.

2. Bias

Key point: Bias is a model's inability to capture the true relationship. A high-bias model makes the same kind of mistake whatever data it is trained on.

Suppose the true relationship between input and output is a curve, and we fit a straight line. However the line is placed, it cannot bend to follow the curve. The error that comes from the model's shape being too simple is its **bias**.

A high-bias model:

- has a large error on the **training data** itself, because it cannot even fit the points it learns from;
- also has a large error on new data;
- is **underfitting**.

3. Variance

Key point: Variance is how much a model changes when trained on a different sample of data. A high-variance model fits the noise of whichever sample it saw.

Now fit a very flexible model, such as a high-degree polynomial. It can bend through almost every training point, so its training error is tiny. But train it again on a different random

sample from the same source, and it bends differently, because it followed the noise of each sample.

How much the model's predictions change from one training set to another is its **variance**. A high-variance model:

- has a small error on the training data;
- has a large error on new data, because it learned noise that does not repeat;
- shows a big gap between training and test performance;
- is **overfitting**.

This is a different meaning from the variance of a column, the average squared distance of its values from their mean (see the [understanding your data Note](#), section seven point one, and the [PCA geometric intuition Note](#), section five). Model variance is the spread of the model's predictions across training sets, not the spread of the data.

4. Seeing bias and variance

Key point: Train the same model on many training sets. The spread of the resulting curves is the variance; how far their average is from the truth is the bias.

Figure 1 makes this concrete. The true relationship is a wave (dashed black). From it we draw 200 different training sets of 40 noisy points, train a model on each, and draw 20 of the resulting curves (orange) and their average (blue).

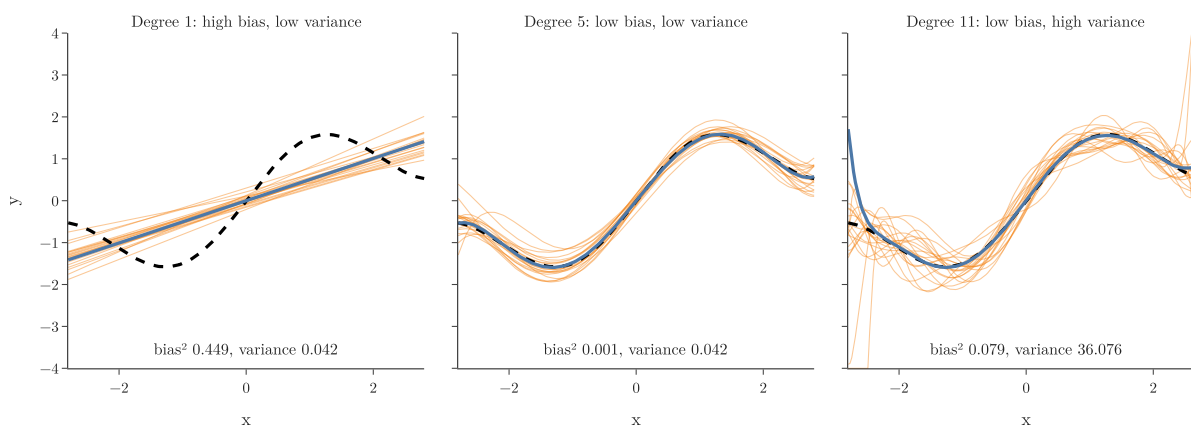


Figure 1: The same model trained on 20 different training sets

Model	Curves	Bias ²	Variance
Degree 1	all similar, all wrong	0.449	0.042
Degree 5	similar and close to the truth	0.001	0.042
Degree 11	close on average, wildly different	0.079	36.1

- **Degree 1** (a straight line) gives nearly the same line every time: low variance. But that line can never follow the wave: high bias.
- **Degree 11** follows the wave on average (low bias in the middle) but every curve swings differently, wildly so near the edges: very high variance.
- **Degree 5** has both low: the target.

The dartboard in Figure 2 is a common way to picture the four combinations: bias is how far the throws land from the centre on average, variance is how scattered they are.

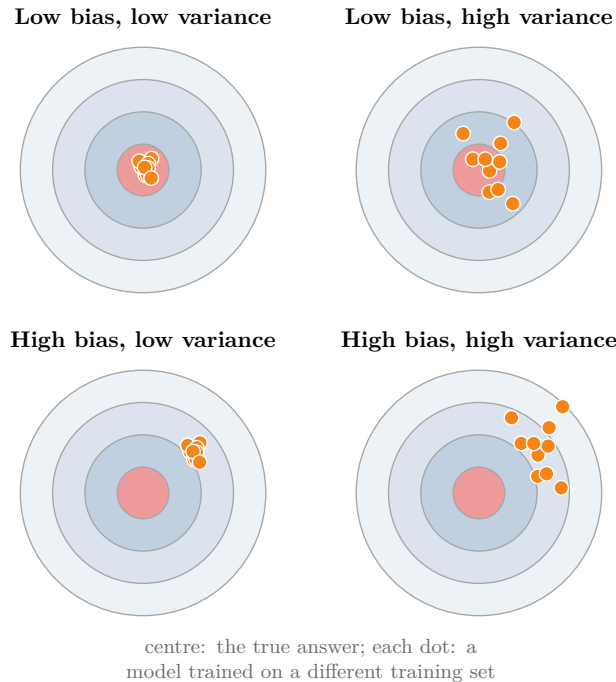


Figure 2: Bias and variance as a dartboard

5. The trade-off

Key point: As a model becomes more complex, bias falls and variance rises. The best model is at the complexity where their sum is smallest.

The expected squared error of a model on new data splits into three parts:

$$\text{expected error} = \text{bias}^2 + \text{variance} + \text{noise}$$

- **Bias²:** error from the model being too simple.
- **Variance:** error from the model depending too much on the particular training set.
- **Noise:** randomness in the data itself. No model can remove it.

With numbers, for degree 5: $0.001 + 0.042 + 0.25 = 0.293$. For degree 1: $0.449 + 0.042 + 0.25 = 0.741$.

Figure 3 measures all three for degrees 1 to 11.

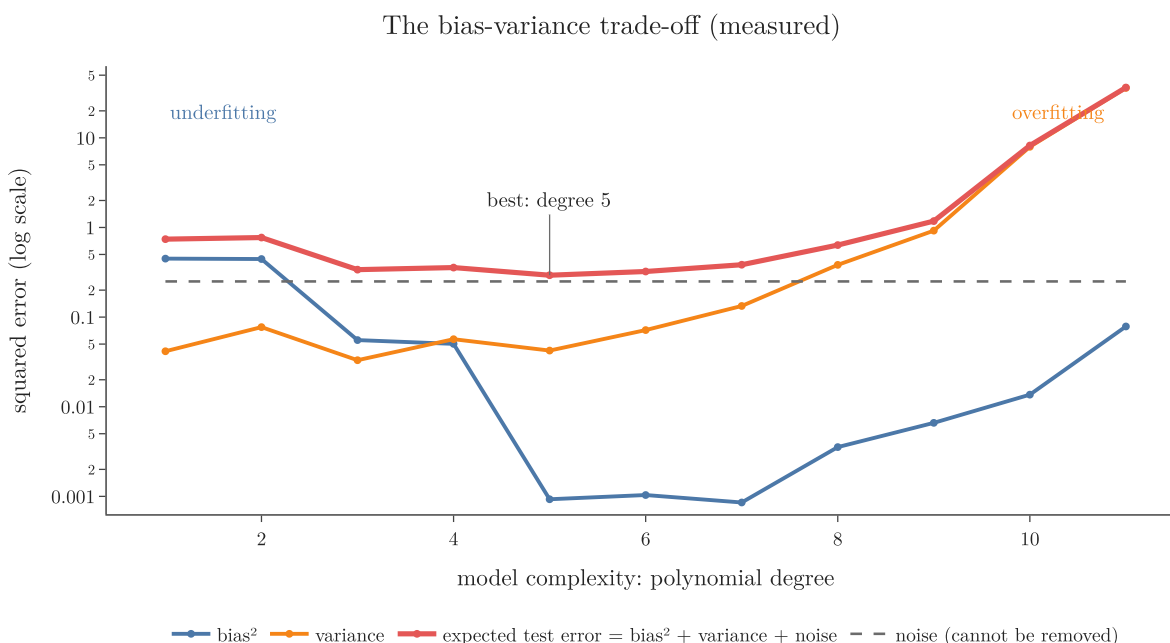


Figure 3: Bias², variance and expected test error against the degree

- On the left, simple models have high bias and low variance: **underfitting**.
- On the right, complex models have low bias and high variance: **overfitting**.
- The total error is smallest in between, here at degree 5, close to the noise floor of 0.25.

Making a model more flexible buys lower bias at the price of higher variance, and the reverse. That is the trade-off. The goal is not zero bias or zero variance, but the lowest total error.

Extra: In Figure 3, bias^2 creeps up again at the highest degrees. That is a side effect of the wild swings near the edges of the data, which also move the average curve. The textbook picture, in which bias falls steadily with complexity, holds for the middle of the data.

6. What to do about it

Key point: Underfitting: make the model more flexible or add better features. Overfitting: get more data, simplify the model, or use regularisation, bagging or boosting.

Problem	Signs	Remedies
High bias (underfitting)	high training error, test error about the same	a more complex model, more or better features (feature engineering)

Problem	Signs	Remedies
High variance (overfitting)	low training error, much higher test error	more training data, a simpler model, regularisation, bagging

Three standard techniques, all covered later, target this trade-off directly:

- **Regularisation** (Ridge, Lasso, Elastic Net, the next Notes): keeps a flexible model but penalises large coefficients, lowering variance for a small increase in bias.
- **Bagging** (for example random forests): averages many high-variance models trained on different samples, which cancels much of the variance.
- **Boosting**: builds a strong model from many simple, high-bias ones, lowering bias step by step.

Extra: The terms come from statistics. The “bias” here is different from the bias (intercept) term of a model and from social bias in data; it means systematic error of the model’s predictions.

7. Summary

	High bias	High variance
Model	too simple	too flexible
Training error	high	low
Test error	high	high
Name	underfitting	overfitting
Fixes	more complex model, better features	more data, simpler model, regularisation, bagging

- Expected error = $\text{bias}^2 + \text{variance} + \text{noise}$; the noise cannot be removed.
- More complexity lowers bias and raises variance; the best model balances them.
- On the wave example, degree 5 gives the lowest total error (0.29), against 0.74 for a line and 36 for degree 11.

8. Key terms

Term	Meaning
Bias	Error from a model being too simple to capture the true relationship
Variance	How much a model's predictions change when it is trained on a different sample of data
Bias-variance trade-off	Lowering bias by adding complexity tends to raise variance, and the reverse
Noise (irreducible error)	Randomness in the data that no model can predict
Regularisation	Penalising large coefficients to reduce a model's variance
Bagging	Averaging many models trained on different samples of the data to reduce variance
Boosting	Combining many simple models in sequence to reduce bias